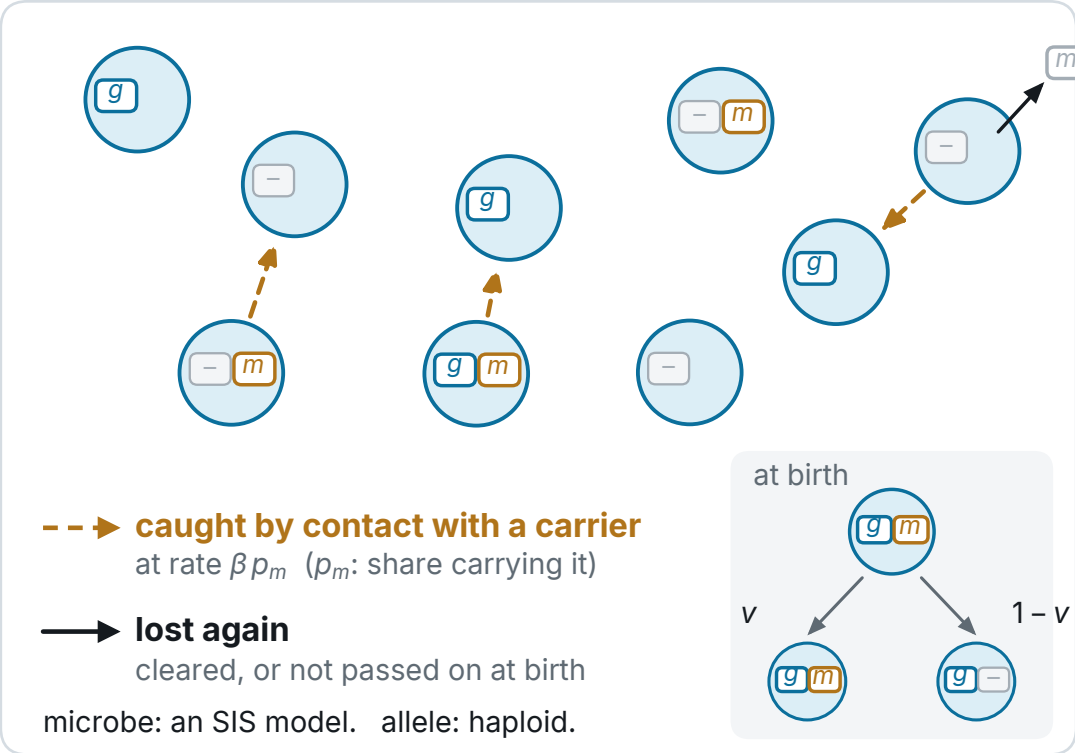
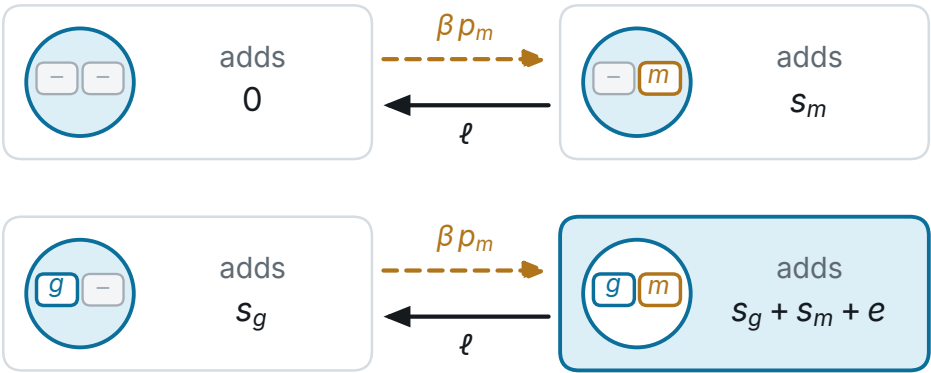


A host allele that is inherited, a microbe that is caught and lost



Four kinds of host



what each kind adds to a growth rate that is otherwise negative

host fitness = $\underbrace{g}_{s_g \text{ the allele}} + \underbrace{m}_{s_m \text{ direct effect}} + \underbrace{g \times m}_e \text{ interaction}$

the microbe can help directly, or only together with the allele